

Chapter 9

Relations

Relations and Their Properties

Section 9.1

Section Summary

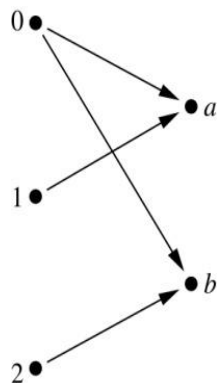
- Relations and Functions
- Properties of Relations
 - *Reflexive Relations*
 - *Symmetric and Antisymmetric Relations*
 - *Transitive Relations*
- Combining Relations

Binary Relations

Definition: A *binary relation* R from a set A to a set B is a subset $R \subseteq A \times B$.

Example:

- Let $A = \{0, 1, 2\}$ and $B = \{a, b\}$
- $\{(0, a), (0, b), (1, a), (2, b)\}$ is a relation from A to B .
- We can represent relations from a set A to a set B graphically or using a table:



R	a	b
0	×	×
1	×	
2		×

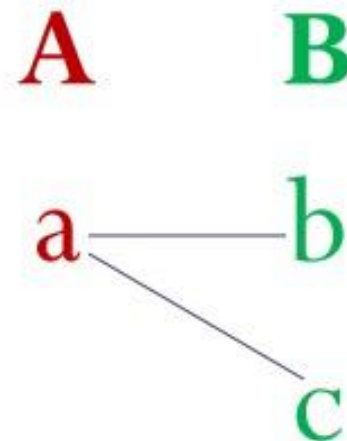
Relations are **more general** than functions. A function is a relation where **exactly one element** of B is related to each element of A .

Relations and their properties

- **Sets** of ordered pairs are called **Binary relations**

A **B** **A** **B**

- E.g. $R = \{\{a, b\}, \{a, c\}\}$
- **Binary relations** represent relationship between the elements of two sets.
- Let **A**, **B** be any **two sets**.



- **R** subset **AxB**

Relations and their properties

- A *binary relation* R from A to B , written $R:A \leftrightarrow B$, is a **subset** of the **Cartesian product** of the two sets $A \times B$.
- In other words, a **binary relation** from A to B is a **set** R of **ordered pairs** where the **first element** of each ordered pair comes from A and the **second element** comes from B .

Relations and their properties

- The notation $a R b$ means $(a,b) \in R$.
- The notation $a \cancel{R} b$ means $(a,b) \notin R$.
- If $a R b$ we may say “ a is *related* to b (*by relation* R)”, or
- “ a *relates* to b (*under relation* R)”.

Relations Representation

```
graph TD; A[Relations Representation] --> B[Roster Notation]; A --> C[Builder notation]; A --> D[Graph]; A --> E[Table];
```

**Roster
Notation**

**Builder
notation**

Graph

Table

Roster notation

- **Roster notation** is a list of elements, separated by commas, enclosed in curly braces.
- Example: $\{3, 5, 7\}$ is the set of single-digit odd prime numbers.
- Example: $A = \{1, 2, 3, 4, 5\}$

Builder notation

Builder notation gives a rule to **follow** that will tell you **how to build the roster**

$A = \{x \mid 0 < x < 6, x \text{ is a whole number}\}$

Relations can be represented by:

A. Roster Notation:

List of ordered pairs

Ex: the set $\{1,2,3,4\}$

B. Set builder notation

Ex: $R = \{(a,b) \mid a \text{ divides } b\}$

Relations can be represented by:

Ex: Let A be the set $\{1,2,3,4\}$, which ordered pairs are in the relation $R=\{(a,b) \mid a \text{ divides } b\}$?



Graph notation

$R1 = \{(1,1), (2,2), (3,3), (4,4), (1,3), (1,2), (1,4), (2,4)\}$

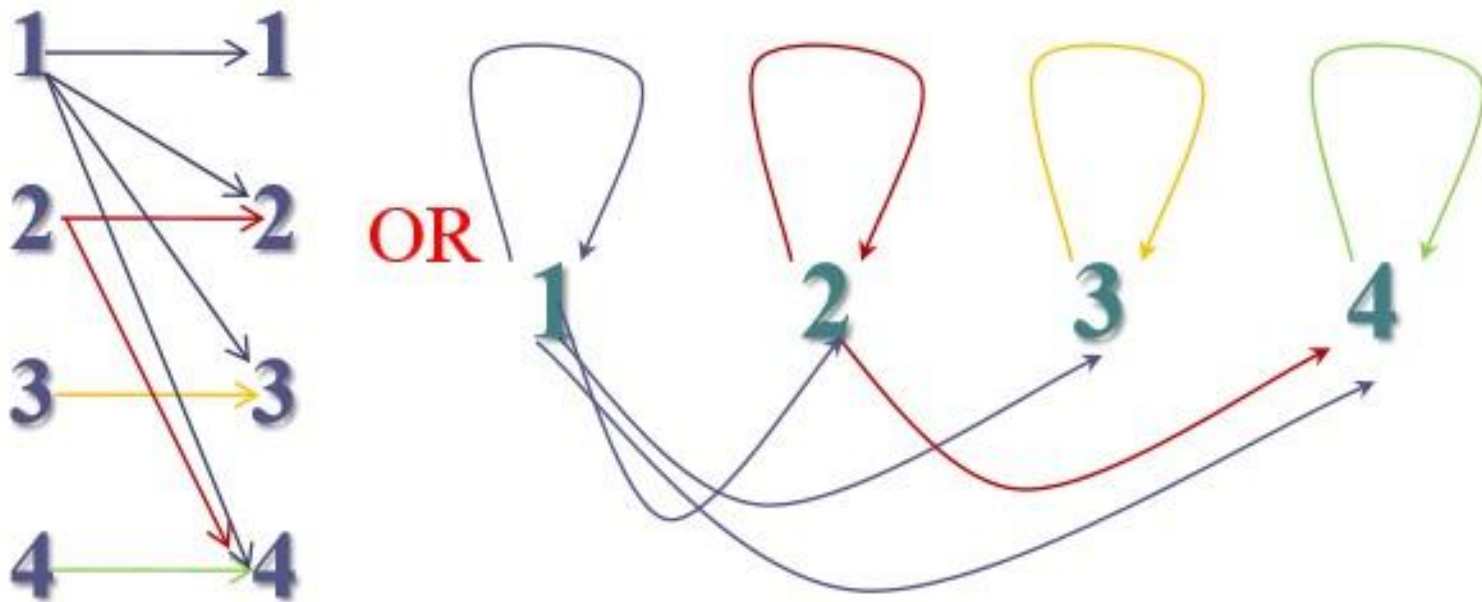


Table notation

$$R = \{(1,1), (2,2), (3,3), (4,4), (1,3), (1,2), (1,4), (2,4)\}$$

R	1	2	3	4
1	×	×	×	×
2		×		×
3			×	
4				×

Example

Let $R: A \rightarrow B$

$A = \{1, 2, 3\}$ represents
students

$B = \{a, b\}$ represents
courses

Example

$$A \times B =$$

$$\{ (1,a), (1,b), (2,a), (2,b), (3,a), (3,b) \}$$

$R = \{ (1,a), (1,b) \} \Rightarrow$ it means
that **student 1** registered in
courses a and b

Relations

Example: $R: A \rightarrow B$ such that $(a > b)$

$$A = \{1, 2, 3, 4\} \quad B = \{2, 3, 4\}$$

Solution:

$$\{(3, 2), (4, 2), (4, 3)\}$$

Relations on Set

A (binary) relation from a set A to itself is called a *relation on* the set A .

Example

- Consider these relations on the set of integers:
- $R1 = \{(a,b) | a \leq b\},$
- $R2 = \{(a,b) | a > b\},$
- $R3 = \{(a,b) | a = b \text{ or } a = -b\},$
- $R4 = \{(a,b) | a = b\},$
- $R5 = \{(a,b) | a = b + 1\},$
- $R6 = \{(a,b) | a + b \leq 3\}.$

Which of these relations contain each of the pairs $(1, 1), (1, 2), (2, 1), (1, -1),$ and $(2, 2)$?

Example

Solution:

- $(1,1)$ R1, R3, R4, R6
- $(1,2)$ R1, R6
- $(2,1)$ R2, R5, R6
- $(1,-1)$ R2, R3, R6
- $(2,2)$ R1, R3, R4

Identity relation I_A ε

The *identity relation* I_A on a set A is the set $\{(a,a) | a \in A\}$.

Example:

$$A = \{1,2,3,4\}$$

$$I_A = \{(1,1), (2,2), (3,3), (4,4)\}$$

Properties of Relations

Reflexive

if $(a, a) \in R$

Symmetric

if $(a, b) \in R \leftrightarrow (b, a) \in R$. where $a, b \in A$

Transitive

Iff
 $(a, b) \in R \wedge (b, c) \in R$
 $\rightarrow (a, c) \in R.$

1. Reflexivity & Irreflexivity

A relation R on A is *reflexive* if $(a,a) \in R$ for every element $a \in A$.

Ex: Consider the following relations on $\{1,2,3,4\}$

1. $R_1 = \{(1,1), (1,2), (2,1), (2,2), (3,4), (4,1), (4,4)\}$

2. $R_2 = \{(1,1), (2,1), (2,2), (3,3), (3,4), (4,4)\}$

3. $R_3 = \{(a, b) \mid a \leq b\}$

1. Reflexivity & Irreflexivity – cont.

Reflexive

because $R_3 = \{(\mathbf{1},\mathbf{1}), (1,2), (1,3), (1,4),$
 $(\mathbf{2},\mathbf{2}), (2,3), (2,4),$
 $(\mathbf{3},\mathbf{3}), (3,4),$
 $(\mathbf{4},\mathbf{4})\}$

Difference between Not Reflexive and Irreflexive

A relation R on A is **irreflexive** if
for every element $a \in A$,

$$(a, a) \notin R$$

Note: “*irreflexive*” $\neq$ “*not reflexive*”

Difference between Not Reflexive and Irreflexive

Example:

$$A = \{1, 2\}$$

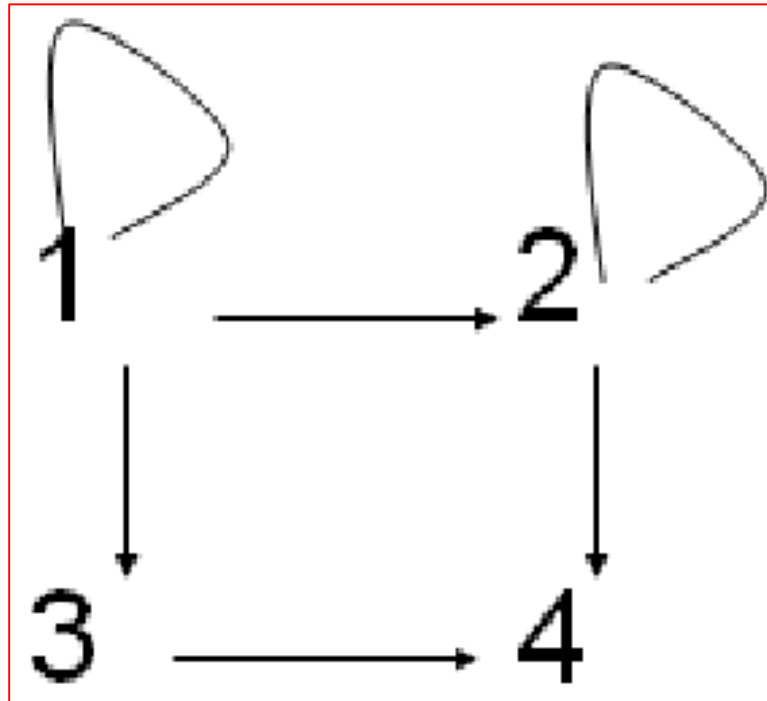
$$R = \{(1, 2), (2, 1), (1, 1)\}$$



Note: “irreflexive” $\neq$ “not reflexive”

Reflexive or not?

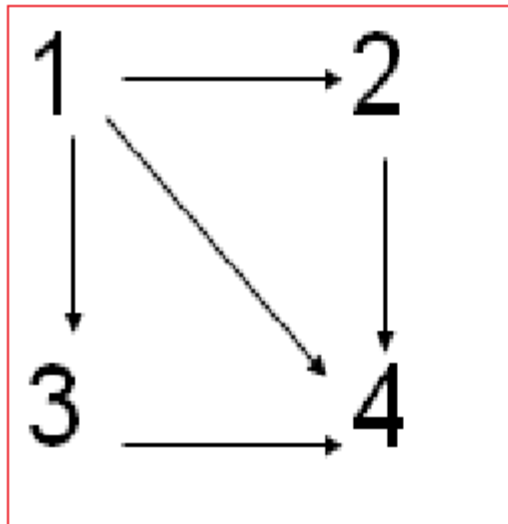
Irreflexive or not?



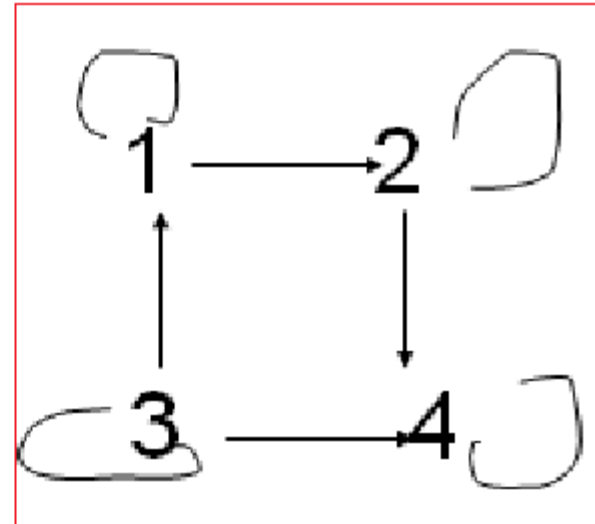
Not Reflexive and Not Irreflexive

Reflexive or not?

Irreflexive or not?



Irreflexive



Reflexive

2. Symmetry & Antisymmetry

A binary relation R on A is **symmetric** if $(\underline{a.b}) \in R \leftrightarrow (\underline{b.a}) \in R$. where $a, b \in A$

A binary relation R on A is **antisymmetric** if $(\underline{a.b}) \in R \rightarrow (\underline{b.a}) \notin R$.

also: $(\underline{a.b}) \in R \wedge (\underline{b.a}) \in R \rightarrow (a=b)$

2. Symmetry & Antisymmetry

Symmetric

When (a,b) is there, then the other pair (b,a) must be there

Antisymmetric

If (a,b) is there, then (b,a) shouldn't be there. Also when $a=b$ then it is antisymmetric

Reflexive or not Irreflexive or not Symmetric or not Antisymmetric or not?

Let $A = \{1, 2, 3\}$

$R1 = \{(1, 2), (2, 2), (3, 1), (1, 3)\}$

not reflexive,
not irreflexive,
not symmetric,
not antisymmetric

Reflexive or not Irreflexive or not Symmetric or not Antisymmetric or not?

Let $A = \{1,2,3\}$

$R_2 = \{(2,2), (1,3), (3,2)\}$	not reflexive, not irreflexive, not symmetric, Antisymmetric
$R_3 = \{(1,1), (2,2), (3,3)\}$	reflexive, not irreflexive, symmetric, antisymmetric

Reflexive or not Irreflexive or not Symmetric or not Antisymmetric or not?

Let $A = \{1,2,3\}$

$R_4 = \{(2,3)\}$

not reflexive,
irreflexive,
not Symmetric,
antisymmetric

Symmetric or not

Antisymmetric or not?

Consider these relations on the set of integers:

$$R1 = \{(a, b) \mid a=b\}$$

symmetric , antisymmetric

$$R2 = \{(a, b) \mid a>b\}$$

not symmetric, antisymmetric

$$R3 = \{(a, b) \mid a=b+1\}$$

not symmetric, antisymmetric

Symmetric or not

Antisymmetric or not?

Note:

- The terms **symmetric** and **antisymmetric** are **not opposites**, because a relation can have both of these properties or may lack both of them.
- A relation **cannot be both symmetric and antisymmetric** if it contains some pair of the form (a, b) , where **$a \neq b$**

3. Transitivity

- A relation R is *transitive* iff (for all a, b, c)
 $(a, b) \in R \wedge (b, c) \in R \rightarrow (a, c) \in R.$

Examples:

$A = \{1, 2\}$

$R_1 = \{(1, 1), (1, 2), (2, 1), (2, 2)\}$

transitive

$R_2 = \{(1, 1), (1, 2), (2, 1)\}$ not transitive,
 $(2, 2) \notin R_2$

Properties of Relations

Iff
 $(a,b) \in R \wedge (b,c) \in R \rightarrow (a,c) \in R.$

if $(a,a) \in R$

if $(a,b) \in R \leftrightarrow (b,a) \in R$ where $a, b \in A$

Reflexive

Example: $A = \{1, 2\}$

$R = \{(1, 2), (2, 1), (1, 1)\}$

1. If all (a,a) are in R then it is **reflexive**
2. If one or more (not all) (a,a) in R are missing then the relation is **not reflexive**
 • **Not Reflexive** because $(2, 2)$ is not there
3. If no (a,a) are in R then the relation is **irreflexive**.
4. If one of the (a,a) exist in R then the relation is **not irreflexive**
 • **Not irreflexive** because $(1,1) \in R$

Symmetric

Symmetric

When (a,b) is there, then the other pair (b,a) must be there

Antisymmetric

If (a,b) is there, then (b,a) shouldn't be there. Also when $a=b$ then it is antisymmetric

$(a,b) \in R \rightarrow (b,a) \notin R.$

also: $(a,b) \in R \wedge (b,a) \in R \rightarrow (a=b)$

Transitive

A relation R is transitive iff (for all a,b,c)
 $(a,b) \in R \wedge (b,c) \in R \rightarrow (a,c) \in R.$

Combining Relations

Let $A = \{1, 2, 3\}$, $B = \{1, 2, 3, 4\}$

$R1 = \{(1, 1), (2, 2), (3, 3)\}$

$R2 = \{(1, 1), (1, 2), (1, 3), (1, 4)\}$

$R1 \cup R2 = \{(1, 1), (2, 2), (3, 3),$
 $(1, 2), (1, 3), (1, 4)\}$

$R1 \cap R2 = \{(1, 1)\}$

$R1 - R2 = \{(2, 2), (3, 3)\}$

$R2 - R1 = \{(1, 2), (1, 3), (1, 4)\}$

Combining Relations

Remember

For sets A , B , the *difference* of A and B , written $A - B$, is the set of all **elements** that are **in A but not in B**.

$$A - B \equiv \{x \mid x \in A \wedge x \notin B\}$$

$$\text{Ex: } \{1, 2, 3, 4, 5, 6\} - \{2, 3, 5, 7, 9, 11\} = \{1, 4, 6\}$$

Composition

Definition: Suppose

- R_1 is a relation from a set A to a set B .
- R_2 is a relation from B to a set C .

Then the *composition* (or *composite*) of R_2 with R_1 , is a relation from A to C where

- if (x,y) is a member of R_1 and (y,z) is a member of R_2 , then (x,z) is a member of $R_2 \circ R_1$.
- If (a,c) is in R_1 and (c,b) is in R_2 then (a,b) is in $R_2 \circ R_1$

Composite Relations

Ex: R is the relation from $\{1,2,3\}$ to $\{1,2,3,4\}$ $R = \{(1,1), (1,4), (2,3), (3,1), (3,4)\}$

S is the relation from $\{1,2,3,4\}$ to $\{0,1,2\}$

$S = \{(1,0), (2,0), (3,1), (3,2), (4,1)\}$

- $S \circ R = \{(1,0), (1,1), (2,1), (2,2), (3,0), (3,1)\}$

The range for first relation is the domain for second one.

Example

Find $S \circ R$ if $R = \{(1,1), (1,2), (2,5), (3,4)\}$ and $S = \{(1,6), (2,1), (4,0)\}$

$S \circ R = \{(1,6), (1,1), (3,0)\}$

Power

- Let R be a relation on the set A .
the power R^n , $n=1,2,3\dots$ are defined by

$$R^1 = R \quad \text{and} \quad R^{n+1} = R^n \circ R$$

Ex: Let $R = \{(1,1), (2,1), (3,2), (4,3)\}$.

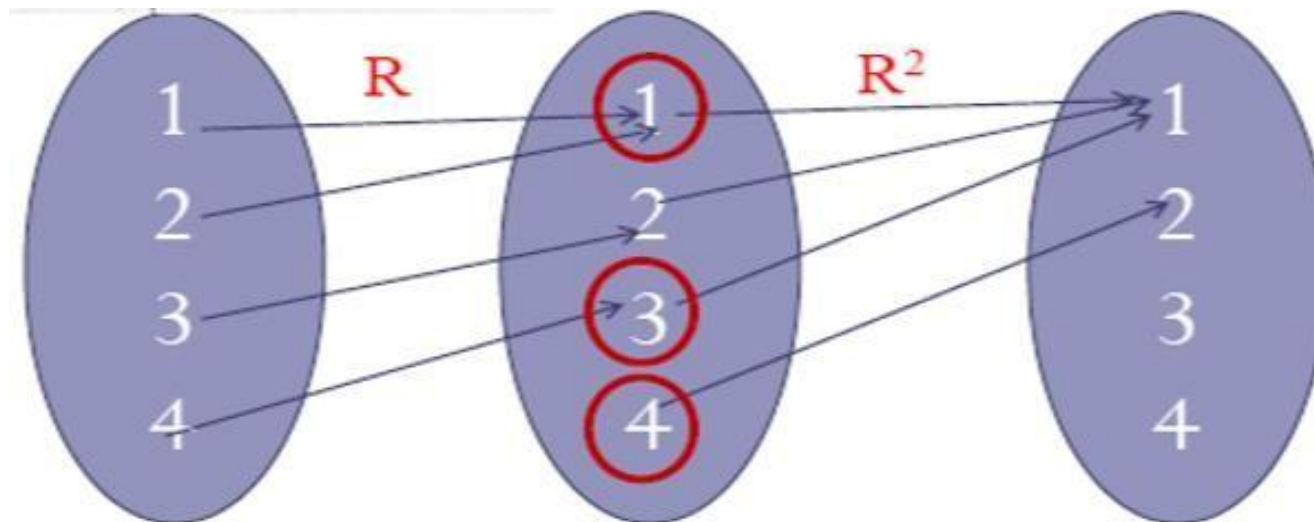
Find:

- $R^2 = R \circ R = \{(1,1), (2,1), (3,1), (4,2)\}$
- $R^3 = R^2 \circ R = \{(1,1), (2,1), (3,1), (4,1)\}$

Power

- Additional computations shows that R^4 is the same as R^3 , so $R^4 = R^3 \circ R = \{(1,1), (2,1), (3,1), (4,1)\}$
- It also follows that $R^n = R^3$ for $n=5,6,7,\dots$. Please verify this!

Example



$$R^3 = \{(1,1), (2,1), (3,1), (4,1)\}$$

Representing Relations

Section 9.3

Section Summary

- Representing Relations using Matrices
- Representing Relations using Digraphs

Representing Relations

Some special ways to represent binary relations:

- With **ordered pairs** (section 9.1)
- With **a table** (section 9.1)
- With a **Zero-one matrix**
- With a **directed graph** or **diagraph**

Using Zero-One Matrices

- To represent a **relation R** by a **matrix $M_R = [m_{ij}]$** , let $m_{ij} = 1$ if $(a_i, b_j) \in R$, else 0.
- *E.g.*, **Joe** likes **Susan** and **Mary**, **Fred** likes **Mary**, and **Mark** likes **Sally**.

Using Zero-One Matrices

- The **0-1 matrix** representation of that “**Likes**” relation:

	Susan	Mary	Sally
Joe	1	1	0
Fred	0	1	0
Mark	0	0	1

Using Zero-One Matrices

Example:

$A = \{1, 2, 3\}$, $B = \{1, 2\}$, $R: A \leftrightarrow B$ such that:

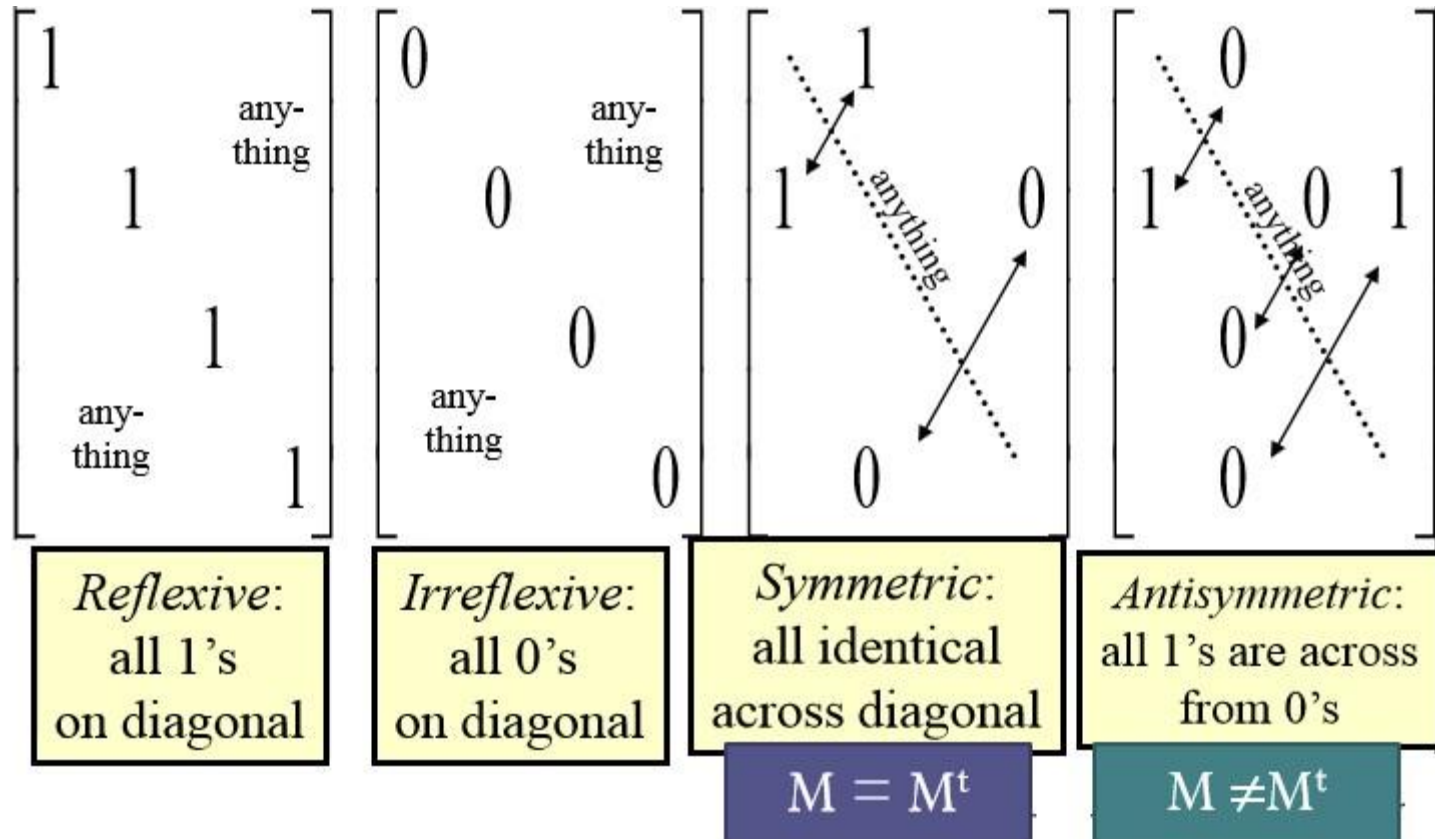
$R = \{(2, 1), (3, 1), (3, 2)\}$ ($a > b$) then the
matrix for R is:

$$M_R = \begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 1 & 1 \end{bmatrix}$$

Zero-One Reflexive, Symmetric

- Terms:
- ***Reflexive and Irreflexive, symmetric and antisymmetric.***
 - These relation characteristics are very easy to recognize by inspection of the zero-one matrix.

Zero-One Reflexive, Symmetric



Example

Is **R** reflexive, symmetric, antisymmetric?

$$M_R = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

Reflexive, symmetric, not antisymmetric

The Boolean Operations 1) join and meet

1) Join and meet representing Union and the Intersection

The Boolean Operations **join** $\vee$ and **meet** $\wedge$ can be used $\rightarrow$ to find the matrices representing the **union** $\cup$ and the **intersection** $\cap$ of two relations

Then:

$$M_{R1 \cup R2} = M_{R1} \vee M_{R2}$$

$$M_{R1 \cap R2} = M_{R1} \wedge M_{R2}$$

Example

Suppose that the relations **R1** and **R2** on set A are represented by the matrices:

$$M_{R1} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \quad \text{and} \quad M_{R2} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$

What are the **matrices**
representing $R1 \cup R2$ and $R1 \cap R2$

Example

$$\begin{aligned} M_{R1 \cup R2} &= M_{R1} \vee M_{R2} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix} \\ M_{R1 \cap R2} &= M_{R1} \wedge M_{R2} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \end{aligned}$$

2) Composite

2) Composite

Suppose that $\mathbf{R}: A \leftrightarrow \mathbf{B}$, $\mathbf{S}: \mathbf{B} \leftrightarrow C$

(Boolean Product)



$$M_{S \circ R} = M_R \odot M_S$$

Remember how we did Boolean product!!!!!!

$$\bullet A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$

Remember how we did Boolean product!!!!!!

$A \odot B$

=

$$\begin{bmatrix} (1 \wedge 1) \vee (0 \wedge 0) & (1 \wedge 1) \vee (0 \wedge 1) & (1 \wedge 0) \vee (0 \wedge 1) \\ (0 \wedge 1) \vee (1 \wedge 0) & (0 \wedge 1) \vee (1 \wedge 1) & (0 \wedge 0) \vee (1 \wedge 1) \\ (1 \wedge 1) \vee (0 \wedge 0) & (1 \wedge 1) \vee (0 \wedge 1) & (1 \wedge 0) \vee (0 \wedge 1) \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

Example

$$M_{S \circ R} = M_R \ominus M_S \neq M_S \ominus M_R$$

Let:

$$M_R = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \text{and} \quad M_S = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix}$$

Find the matrix of $S \circ R$?

Example

$$M_{S \circ R} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

The *boolean product* of **A** and **B** is **like** normal **matrix multiplication**, But using $\vee$ instead $+$
And using $\wedge$ instead $\times$

Attention

$$M_{S \circ R} = M_R \Theta M_S \neq M_S \Theta M_R$$

Check!

$$M_S = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix} \quad \text{and} \quad M_R = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Attention

$$M_{S \circ R} = M_S \Theta M_R = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

$$\neq M_R \Theta M_S = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

3) Power

$$M_{R^n} = M_R^{[n]}$$

- $R^2 = R \circ R = M_R^{[2]}$
- $R^3 = R^2 \circ R = M_R^{[3]}$

Example

- Find the matrix that represent $\mathbf{R}^2$ where the matrix representing $\mathbf{R}$ is:

$$M_R = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix} \quad \text{then} \quad M_{R^2} = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

Example

$$M_R = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix} \quad \text{and} \quad M_{R^2} = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

- Find the matrix that represent R^3 where the matrix representing R and R^2 is:

$$M_{R^3} = M_{R^2} \Theta M_R = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$

Example

$$M_R = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix} \quad \text{and} \quad M_{R^2} = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

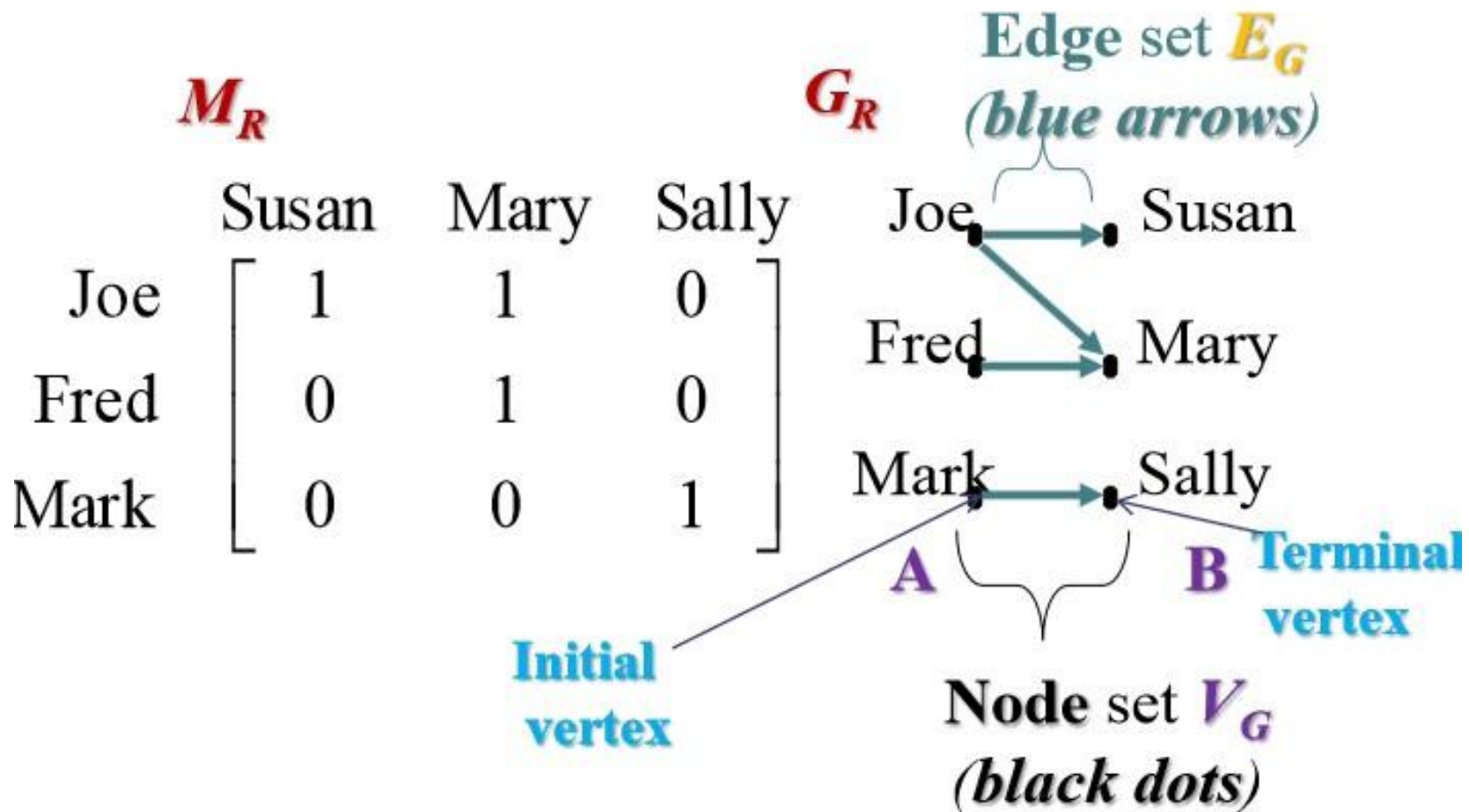
$$= M_R \ominus M_{R^2} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$M_{R^3} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

Using Directed Graphs

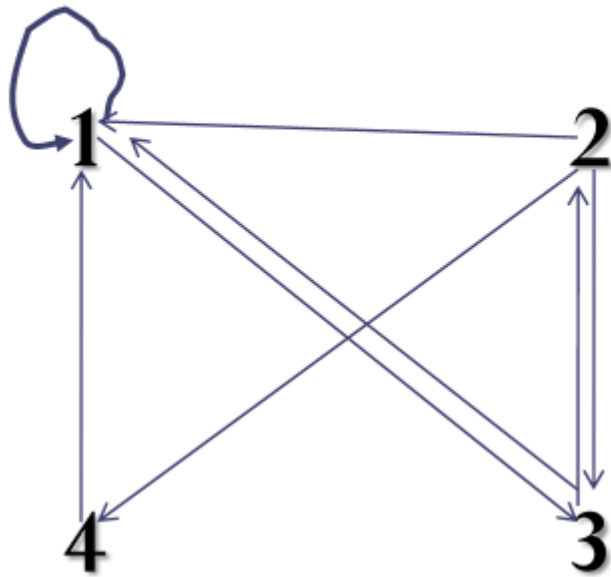
- A directed graph or digraph $G=(V_G, E_G)$ is a set V_G of **vertices** (**nodes**) with a set $E_G \subseteq V_G \times V_G$ of **edges** (**arcs**, **links**).
- Visually represented using **dots** for **nodes**, and **arrows** for **edges**.
- Notice that a **relation** $R:A \leftrightarrow B$ can be represented as a **graph** $G_R=(V_G=A \cup B, E_G=R)$.

Using Directed Graphs



Example

- The directed graph of the relation
- $R = \{(1,1), (1,3), (2,1), (2,3), (2,4), (3,1), (3,2), (4,1)\}$
on the set $\{1,2,3,4\}$ is



Directed graph and relation properties

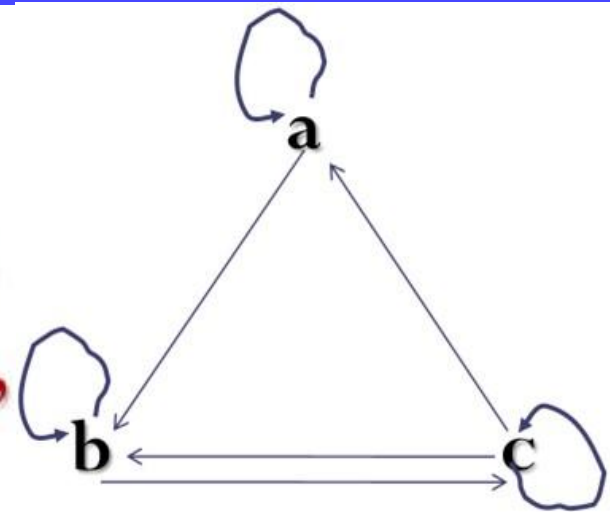
- The **directed graph** of the **relation** can be used to determine whether the relation has various properties.
- **A relation is reflexive**
- **If and only if** there is a **loop** at every **vertex** of the directed graph, so that every ordered pair of the form (x,x) occurs in the relation.
- **A relation is symmetric**
- **If and only if** for **every edge** between distinct vertices in its diagraph **there is an edge in the opposite direction**, so that (y,x) is in the relation whenever (x,y) is in the relation.

Directed graph and relation properties. cont

- A relation is antisymmetric
- If and only if there are **never two edges in opposite directions** between distinct vertices.
- A relation is transitive
- If and only if whenever there is an edge from a vertex **x** to a vertex **y** and an edge from vertex **y** to a vertex **z**, there is an edge from **x** to **z**.
(completing a **triangle** where each side is a directed edge with the correct direction.)

Example

Determine whether the relations for the directed graph are reflexive, symmetric, antisymmetric, and/or transitive



Reflexive: because there are loops at every vertex.

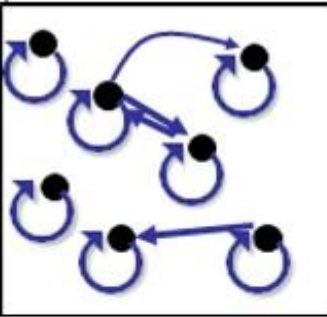
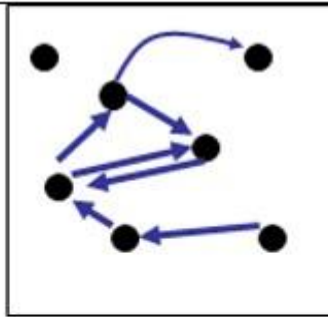
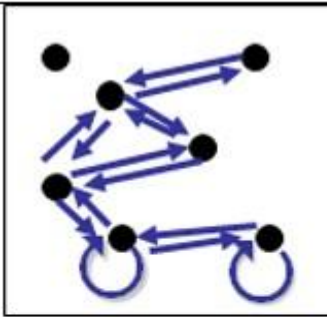
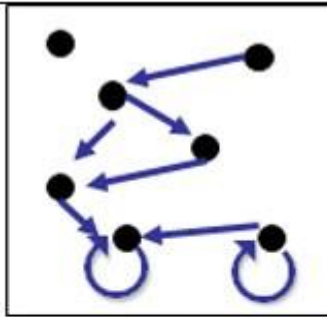
Not symmetric: because there is an edge from **a** to **b** and there is no edge from **b** to **a**.

Not antisymmetric: because there are edges in opposite directions connecting **b** and **c**.

Not transitive: because there is an edge from **a** to **b** and an edge from **b** to **c**, but no edge from **a** to **c**.

Digraph Reflexive, Symmetric

It is extremely easy to recognize the reflexive/irreflexive/symmetric/antisymmetric properties by graph inspection.

			
Reflexive: Every node has a self-loop	Irreflexive: No node links to itself	Symmetric: Every link is bidirectional	Antisymmetric: No link is bidirectional
Not symmetric, non-antisymmetric		Non-reflexive, non-irreflexive	

Closure Relations

Section 9.4

1) Reflexive Closure

- For any property X , the “ **X closure**” of a set A is defined as the “**smallest**” superset of A that has the given property.
- 1) The **reflexive closure** of a relation R on A is obtained
→ by adding **(a,a)** to **R** for each **$a \in A$** *not already in R*

*I.e., it is **$R \cup I_A$***

Example 1

The relation

$R = \{(1,1), (1,2), (2,1), (3,2)\}$ on
the

set $A = \{1,2,3\}$ is not reflexive.

**How can you produce a reflexive
relation containing R that is as
small as possible?**

Answer: By adding $(2,2)$ and
 $(3,3)$ so the :

Reflexive closure of R is:

$\{(1,1), (1,2), (2,1), (3,2), (2,2), (3,3)\}$

Example 2

What is the reflexive closure of the relation:

$R = \{(a, b) \mid a < b\}$ on the set of integers?

Answer: $R \cup \Delta$

The Reflexive Closure of R is:

$$\{(a, b) \mid a < b\} \cup \{(a, a) \mid a \in \mathbb{Z}\} = \{(a, b) \mid a \leq b\}$$

Example:

$$R = \{(a, b) \mid a < b\} = \{(1, 2), (1, 3), (2, 3)\}$$

The Reflexive closure of R =

$$\{(1, 2), (1, 3), (2, 3)\} \cup \{(1, 1), (2, 2), (3, 3)\}$$

2) Symmetric closure

The *symmetric closure* of R is obtained by adding (b, a) to R for each (a, b) in R .

i.e., it is $R \cup R^{-1}$

Example 1

The relation

$\{(1,1), (2,2), (1,2), (3,1), (2,3), (3,2)\}$ on the

set $\{1,2,3\}$ is not symmetric.

How can we produce a symmetric relation that is as small as possible and contains R ?

Answer: by adding $(2,1)$ and $(1,3)$ so the

Symmetric Closure of R is:

$\{(1,1), (2,2), (1,2), (3,1), (2,3), (3,2), (2,1), (1,3)\}$

Example 2

What is the **symmetric closure** of the relation:

$R = \{(a, b) \mid a > b\}$ on the set of positive integers?

Answer:

The Symmetric Closure of R is:

$$\{(a, b) \mid a > b\} \cup \{(b, a) \mid a > b\} = \{(a, b) \mid a \neq b\}$$

Example:

$$R = \{(3, 1), (3, 2), (2, 1)\}$$

$$R^{-1} = \{(1, 3), (2, 3), (1, 2)\}$$

$$R \cup R^{-1}$$

$$= \{(3, 1), (3, 2), (2, 1), (1, 3), (2, 3), (1, 2)\}$$

3) Transitive closure

- The **transitive closure** or **connectivity relation** of R is obtained by repeatedly adding (a, c) to R for each $(a, b), (b, c)$ in R .

• i.e., it is

$$R^* = \bigcup_{n \in \mathbb{Z}^+} R^n$$

- Or in term of zero-one matrices:
- $M_R^* = M_R \vee M_R^{[2]} \vee \dots \vee M_R^{[n]}$

Example 1

$$R^* = R \cup R^2 \cup R^3$$

Ex: $R = \{(1,1), (1,2), (2,1), (3,2)\}$ on the set

$$A = \{1, 2, 3\}$$

- $R^* = R \cup R^2 \cup R^3$

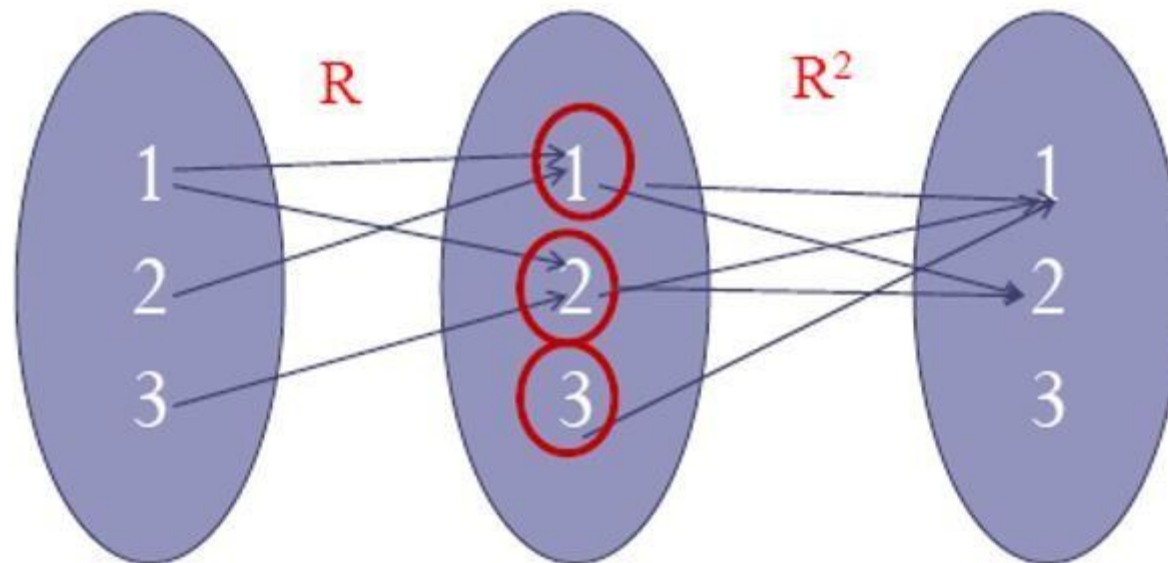
Example 1

$$R^* = R \cup R^2 \cup R^3$$

Ex: $R = \{(1,1), (1,2), (2,1), (3,2)\}$ on the set

$$A = \{1, 2, 3\}$$

$$R^2 = R \circ R = \{(1,1), (1,2), (2,1), (2,2), (3,1)\}$$



$$R^3 = R^2 \circ R$$

$$= \{(1,1), (1,2), (2,1), (2,2), (3,1), (3,2)\}$$

$$R^* = R \cup R^2 \cup R^3$$

$$= \{(1,1), (1,2), (2,1), (2,2), (3,1), (3,2)\}$$

– Or in term of zero-one matrices:

$$- M_R^* = M_R \vee M_{R[2]} \vee \dots \vee M_{R[n]}$$

$$A = \{1, 2, 3\}$$

$$M_R = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

$$M_{R^2} = M_R \ominus M_R$$

$$\begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \ominus \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

$$M_{R^3} = M_{R^2} \Theta M_R$$

– Or in term of zero-one matrices:

– $M_R^* = M_R \vee M_{R[2]} \vee \dots \vee M_{R[n]}$

$$A = \{1, 2, 3\}$$

$$M_R = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \quad R = \{(1,1), (1,2), (2,1), (3,2)\}$$


$$M_{R^*} = M_R \vee M_{R^{[2]}} \vee M_{R^{[3]}} =$$

$$\begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \vee \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \vee \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}$$

$$R^2 = R \circ R =$$

$$\{(1,1), (1,2), (2,1), (2,2), (3,1)\}$$

$$R^3 = R^2 \circ R =$$

$$\{(1,1), (1,2), (2,1), (2,2), (3,1), (3,2)\}$$

$$M_{R^*} =$$

$$\{(1,1), (1,2), (2,1), (2,2), (3,1), (3,2)\}$$

Example 2

- Find M_{R^*} for

$$M_R = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}$$

$$\begin{aligned}
 M_{R^*} &= M_R \vee M_{R^{[2]}} \vee M_{R^{[3]}} = \\
 &\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix} \vee \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} \vee \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} \\
 &= \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}
 \end{aligned}$$

Equivalence Relations

- An equivalence relation on a set A is simply any binary relation on A that is reflexive, symmetric, and transitive.

Example1:

A relation on the set of real numbers such that

$$R = \{(a, b) \mid a - b \text{ is an integer}\}$$

Equivalence Relations

Answer:

R is reflexive since (a,a) is an integer where $a-a=0$

R is symmetric since $a-b$ and $b-a$ is an integer

R is transitive $a-b$ and $b-c$ are integers, therefore $a-b + b-c = a-c$ is also an integer so R is an **Equivalence Relation**.